

Reach-Based Sampling for Multi-Scale Localised Network Centrality

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Abstract

Localised network centrality, closeness and betweenness computed within a distance threshold, underpins comparisons of street importance across places and cities. Exact computation traces shortest routes from every street and becomes expensive at metropolitan scale, which makes sampling appealing: traversals run from a fraction of streets and the results are scaled up. Sampling, however, raises a design problem. Each street’s sampled score is in effect a survey of its own catchment, and catchments vary by orders of magnitude, so a single sampling rate leaves sparse streets with too few respondents while oversampling dense ones; uncertainty then tracks density and undermines comparison across places. We therefore set the sampling rate street by street: a cheap pilot poll measures every catchment, each street’s rate derives from a conservative lower bound on its polled reach so that polling error costs time rather than accuracy, streets the poll cannot certify as populous are enumerated in full, and inverse-probability weighting keeps the estimates unbiased. Where sampling would not reduce cost, the method computes exactly. On four networks, with The Woodlands (TX) held out from all calibration, both metrics preserve the exact ranking ($\rho \geq 0.96$) at 1–20 km; on the held-out network the method routes closeness to exact computation, whereas a fixed rate samples it and misses the accuracy target at 20 km. Including the pilot’s cost, sampled 20 km runs are 5.6–15.5× faster than exact computation. A single tolerance parameter governs precision, and the method is implemented in the open-source `cityseer` package.

Keywords: network centrality, sampling, urban morphology, betweenness, harmonic closeness, computational efficiency

1 Introduction

Urban morphological analysis relies on localised network centrality to reveal patterns of accessibility and flow in street networks (Cooper, 2015), in which nodes represent street junctions. Two measures are central to this work: *closeness* (how easily a node can reach its neighbours) and *betweenness* (how often a node lies on shortest paths between others). Both are computed within distance thresholds that range from pedestrian (≈ 500 m) to metropolitan (≈ 20 km) scales, as implemented in tools such as sDNA (Cooper and Chiaradia, 2020) and `cityseer` (Simons, 2022).

Exact computation requires running a shortest-path traversal from every node, and the cost scales with both the number of nodes and the size of each node’s neighbourhood: on the Greater London network (156,434 nodes, plus a surrounding buffer; Section 2) the exact 20 km betweenness pass alone takes about seven minutes of multi-core compute (430 s; Section 5.1), and costs compound across thresholds and metrics. Sampling offers a way to contain this cost: traversals run from a randomly selected fraction of the nodes, and the results are scaled up. Each traversal visits every node within the threshold of its starting point, so a node’s estimate is assembled from the traversals that reach it.

Sampling raises a difficulty of its own: comparability. Centrality patterns are routinely compared across areas and cities (Strano et al., 2013), and sampled estimates support that

comparison only if they carry comparable uncertainty. Each node’s score is a survey of its own *catchment*, the set of nodes within the distance threshold, and its precision depends on how many sampled sources reach the node. Catchment sizes vary by orders of magnitude between dense cores and sparse peripheries. A fixed rate therefore gives a dense district thousands of reached sources and a cul-de-sac a handful (Figure 1D), so the estimates being compared carry unequal uncertainty. Existing adaptive methods such as KADABRA (Borassi and Natale, 2019) tune their sample counts to a single network-wide accuracy target rather than to the precision of each catchment (Section 6.4).

We therefore measure every catchment with a cheap pilot poll: a small set of traversals runs from random starting points, and each street’s hit count measures the size of its catchment at every distance. Each node is then drawn as a traversal source with a probability sized to its own reach. The sample size follows the standard survey rule for a population of a given size (Eppstein and Wang, 2004), with the catchment rather than the whole network as the population. Section 3 states the full design: inverse-probability weighting for unbiasedness (no systematic over- or under-estimate), a census for small catchments, a work test that selects exact computation where sampling cannot reduce cost, and a shared traversal that serves both metrics from one sample.

Our contributions are:

1. A localised sample-size bound: under a distance threshold the relevant population is the catchment, whose size is the reach r rather than the network size n , so the required samples scale with $\log r$ rather than $\log n$. A whole-network guarantee must hold at all n streets simultaneously, which inflates the required sample count; a per-catchment guarantee covers only the streets of that catchment and needs fewer.
2. A per-node sampling design with a single tolerance parameter ε : a pilot poll measures every catchment, each node receives an inclusion probability sized to a confidence bound on its reach, inverse-probability weighting keeps estimates unbiased, and a per-distance work test selects exact computation where sampling cannot reduce cost.
3. Empirical validation at 1–20 km on four street networks spanning the urban density range, one excluded from every calibration decision, together with an ablation that replaces the measured reach with cheaper stand-ins and shows assumed reach failing exactly where the reach measurements predict.
4. An open-source implementation in the `cityseer` Python package.

Section 2 defines the centrality measures and the evaluation target, Section 3 states the method, Section 4 presents the ablation of the reach measurement, Section 5 reports the validation, and Sections 6 and 7 discuss and conclude.

2 Preliminaries

Both centrality measures are computed with a distance cutoff $d_{\max}$. We write d for $d_{\max}$ when the context is clear.

Harmonic closeness. Closeness measures how easily a node can reach its surroundings (Freeman, 1978). We adopt the *harmonic* form, a sum of inverse distances over reachable nodes, rather than classic closeness, which is ill-defined or unstable on the disconnected, variably sized catchments a distance threshold produces. Harmonic closeness sums bounded per-node contributions, with unreachable nodes contributing zero, so it stays well-defined and comparable

across catchments of different size (Rochat, 2009; Boldi and Vigna, 2014), and its additive form lets a single source’s contribution be estimated independently under sampling (Appendix A):

$$C_H(v; d_{\max}) = \sum_{\substack{u \neq v \\ d(v,u) \leq d_{\max}}} \frac{1}{d(v,u)} \quad (1)$$

Betweenness. Betweenness measures how often a node acts as an intermediary (Freeman, 1977). For each node v , it sums, over all pairs of other nodes within the threshold, the fraction of that pair’s shortest paths that pass through v :

$$C_B(v; d_{\max}) = \sum_{\substack{s \neq t, s \neq v, t \neq v \\ d(s,t) \leq d_{\max}}} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (2)$$

where σ_{st} is the total number of shortest paths from s to t , and $\sigma_{st}(v)$ is the number that pass through v . On undirected graphs each pair of endpoints is counted once (Appendix A gives the ordered-pair weighting). Any piece of a shortest path is itself a shortest path, so a distance-bounded traversal from each source discovers every node that could contribute to betweenness at that threshold; this is what lets the sampling design of Section 3 serve betweenness from the same traversals as closeness. The subpath property holds for shortest-path (metric) distance but need not hold for angular (simplest-path) distance (Turner, 2007); Section 6.3 returns to that case.

Reach. The set of nodes within distance d of a node v is v ’s *catchment*; the *reach* of v is its size, $r_v = |\{u : d(v,u) \leq d\}|$. Reach grows roughly with the square of the analysis distance, and at any fixed distance it scales with local node density, so it varies by orders of magnitude between dense cores and sparse peripheries.

Buffers and live nodes. Nodes near the edge of a network have artificially low centrality because their neighbourhoods are truncated. The standard remedy (Gil, 2017) is to surround the study area with a ring of *buffer nodes*: paths can pass through them, but their own centrality values are not reported. Only the interior (*live*) nodes receive results. We write n_L for the live count, n for the total, and $\varphi = n_L/n$ for the live fraction. A small town buffered to 20 km is mostly buffer, so its live fraction is small; this matters when the cost of exact computation is weighed in Section 3.3.

Accuracy metric. We evaluate sampling quality using Spearman rank correlation (ρ) between sampled and exact centrality values. Urban applications rely on relative rankings (e.g., identifying the most central streets) rather than absolute values, so ρ captures the quantity of practical interest. The coefficient runs from -1 to 1 , and a value of 1 means the sampled ranking matches the exact ranking exactly. We set the target at $\rho \geq 0.95$, a conventional threshold for ranking applications; a looser $\rho = 0.9$ would allow visible reordering among the lower-ranked streets. A validation *cell* is one combination of network, metric, and distance. Two numbers from the sampled 20 km cells make that target concrete: the median street moves by at most 2.5 percentile points in the ranking, and at least 80% of the streets that truly fall in the most-central tenth stay there.

3 Method

The method is a survey. Each street’s score is estimated over its own catchment, so each catchment is a population, and each population needs its own sample size. The population sizes

are unknown, so a cheap poll measures them first; the survey proper is then sized street by street. Three questions organise the section: how many samples a population of a given size needs (Section 3.1), how big each population is (Section 3.2), and when a census is cheaper than a survey (Section 3.3).

3.1 How many samples does a catchment need?

Following Eppstein and Wang (2004), sources are drawn at random, a distance-bounded shortest-path traversal runs from each, and each sampled contribution is scaled up by the reciprocal of its inclusion probability (inverse-probability weighting; Appendix A). The respondents for a target v are its *reached sources*, the sampled sources whose traversals reach v , all of which lie in v 's catchment. Consider first an idealised survey of that catchment, with k sources drawn uniformly at random from the r_v catchment members and each contribution weighted by r_v/k , so that the estimate $\hat{C}_H(v)$ is a sum of k bounded terms. Hoeffding's inequality (Hoeffding, 1963), which bounds how far a sum of independent bounded terms strays from its expectation and thereby determines how many respondents suffice for a stated tolerance, gives

$$\Pr\left[\left|\hat{C}_H(v) - C_H(v)\right| > \varepsilon \cdot C_{\max}\right] \leq 2\exp(-2k\varepsilon^2),$$

where $C_{\max}$ bounds the total centrality attainable over the catchment. Appendix A states the per-term bounds, both for this idealised survey and for the deployed design of Section 3.2; Section 6.2 discusses where the tolerance is loose.

The required k follows from the failure budget δ , the survey's total allowance for streets whose estimates miss the tolerance. Because this allowance is shared among all the streets a guarantee covers, covering more streets leaves each a smaller share and demands more samples: in the global setting of Eppstein and Wang (2004) the budget is divided across all n nodes of the network (a union bound), which puts a $\log n$ factor into k , whereas under localisation only the catchment matters and the budget is divided across the r_v members of v 's own catchment. Solving $2r\exp(-2k\varepsilon^2) \leq \delta$ for k gives the required reached-source count for a catchment of size r :

$$k(r) = \frac{\ln(2r/\delta)}{2\varepsilon^2}. \quad (3)$$

At the default $\varepsilon = 0.05$, a catchment of 40,000 nodes requires roughly 2,700 reached sources. Where reach is locally homogeneous (nearby nodes have catchments of similar size), this allocation carries a catchment-scale analogue of the global guarantee: each of the roughly r_v nodes in a catchment fails with probability at most δ/r_v , so the expected number of tolerance-exceeding nodes per catchment is at most δ . The count scales with $\log r$ rather than $\log n$.

Two parameters enter the bound: the tolerance ε and the failure probability δ . The failure probability is held at its default throughout, so ε is the single calibrated parameter governing precision (Section 4); the work test of Section 3.3 adds one empirically chosen cost constant.

One sample can serve betweenness as well as closeness at almost no extra cost: Brandes' backpropagation (Brandes, 2001) recovers a source's betweenness contributions from the same distance-bounded search tree. The closeness bound does not entail this reuse, so the adequacy of one sample for both metrics is established empirically (Sections 5 and 6.2).

3.2 How big is each population? The pilot poll

The bound of Section 3.1 describes an idealised fixed-size survey of each catchment. The deployed design approximates that survey; this section states the design and returns to the gap between the two. The catchment sizes are unknown, so a pilot polls them. The pilot draws $m = \min(n, \max(400, 0.025n))$ sources uniformly and runs one distance-bounded Dijkstra search (the standard shortest-path algorithm) from each, out to the largest requested threshold. A street with a large catchment is reached by many of these traversals and a street with a small

catchment by few, so each street’s hit count measures its reach at every distance (Appendix A gives the estimator and its confidence bounds). Because the poll travels on the network itself, it respects rivers, railways, dead ends, and disconnected fringes.

The two directions of estimation error carry different costs (Table 1): an overestimated reach lowers the sampling rate and under-samples the catchment, which risks accuracy, whereas an underestimated reach merely oversamples it, which costs time. Inclusion probabilities therefore derive from a one-sided lower confidence bound on the polled reach, a conservative low estimate of the population size, written $\hat{r}_u^-$ (Appendix A):

$$q_u = \min\left(1, \frac{k(\hat{r}_u^-)}{\hat{r}_u^-}\right), \quad (4)$$

floored at $q_u \geq 0.01$ to bound the inverse-probability weights. Figure 1B draws these probabilities on the worked example, with marker size proportional to q . A node the poll rarely hits falls toward $q = 1$: when the poll cannot establish that a population is large, that population is surveyed in full, which is exact for sparse catchments. Contributions are weighted $1/q_u$ per source, so estimates are unbiased whatever the pilot’s quality (Appendix A); pilot error affects precision and running time, not bias.

Table 1: The two directions of pilot error and their costs.

Pilot error	Sampling consequence	Cost
Reach overestimated	Too few respondents	Accuracy at risk
Reach underestimated	Too many respondents	Extra time only
Rates derive from the lower confidence bound on reach, placing the design’s errors on the time side.		

Wherever reach is locally uniform, these probabilities give each target the respondents its reach calls for. A target is answered by the members of its catchment, so its expected count of reached sources is the sum of their probabilities. The lower confidence bound normally sits below the true reach, which raises each q_u , so the expected count meets or exceeds $k(r_v)$ (Figure 1C: 22 reached sources against the required 18). The design falls short only at the boundary of a dense district, where a street is answered mostly by streets whose large catchments earned them low probabilities. Saturation at $q = 1$ around sparse nodes limits that shortfall, and the residual effect is assessed empirically (Section 5.3).

The poll size m affects cost, not correctness: a smaller m widens the confidence bounds, which pushes q toward 1 and the work test (below) toward exact computation, both conservative directions. The pilot parameters (the m rule, the poll level α , the probability floor) are fixed conventions of the implementation; none is fitted to a validation network.

The deployed design still differs from the idealised survey of Section 3.1. Every catchment member is included or not by its own independent draw, so the realised reached-source count varies around its expectation (Figure 1C) and the weights vary across sources. The idealised bound is therefore an approximation for the deployed estimator, assessed empirically (Sections 3.5 and 5).

3.3 When is a census cheaper? The work test

When a population is small, a census is cheaper than a survey. The same holds for a whole distance threshold: at short distances traversals are cheap and sampling saves nothing, so the implementation must decide, per distance, whether to sample at all.

A *call* is one request for a set of metrics and distances. The work test prices exact work separately for each metric, because the required sources differ. Exact *closeness* runs only the n_L live streets: each tallies its own neighbours from its own traversal. Sampled closeness assembles

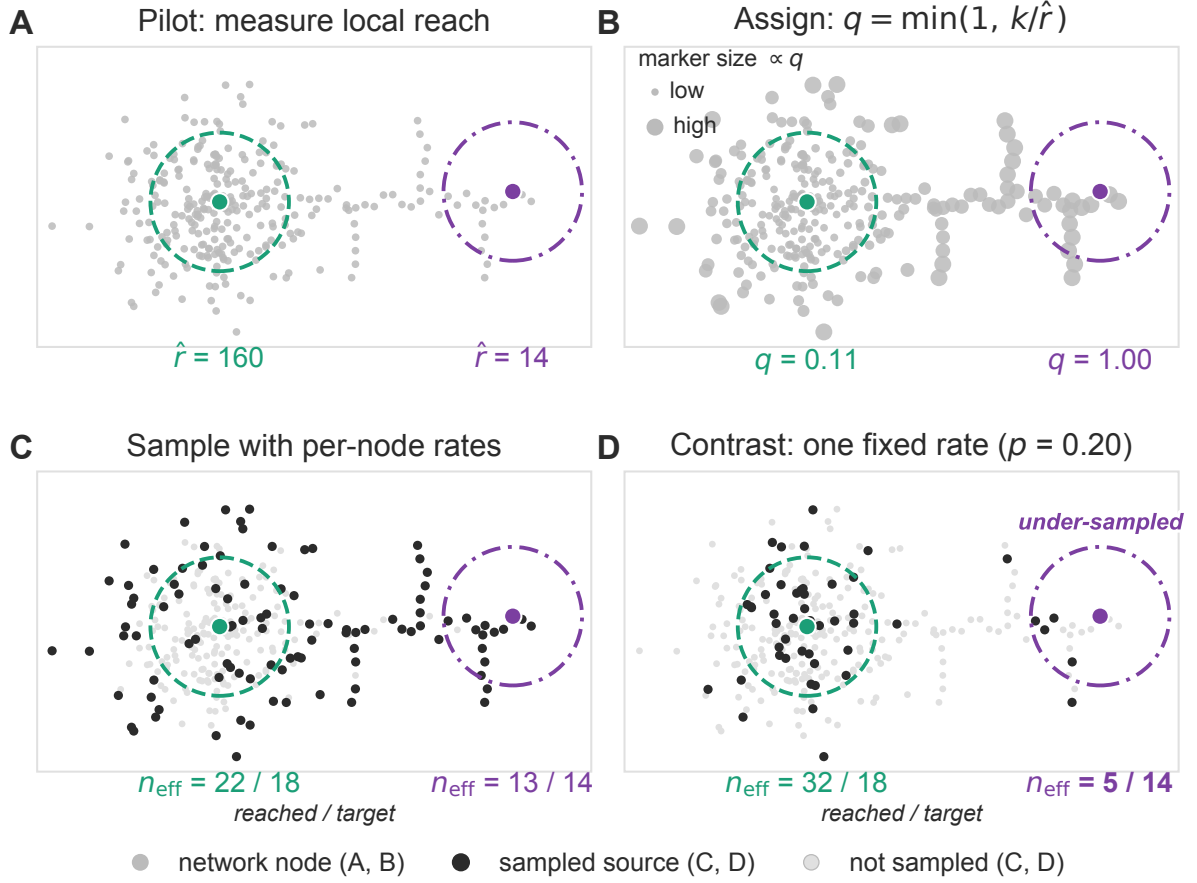


Figure 1: A single sampling rate cannot serve catchments of unequal size; rates sized from measured reach can. The panels illustrate the method on a hypothetical network with a dense core and a sparse fringe, with k fixed at 18 for illustration (in the method k grows slowly with catchment size, Eq. 3). A) The pilot measures each node's catchment; circles mark two exemplar catchments of 160 and 14 nodes. B) Each node receives its own inclusion probability $q = \min(1, k/\hat{r})$, drawn with marker size proportional to q : 0.11 in the core, 1.0 on the fringe. C) One draw under these probabilities: the dense catchment receives 22 reached sources (n_{eff}) against the required 18, and the 14-node catchment is censused (13 of 14 drawn; a few members near the core carry q below 1). Contributions are weighted $1/q$ per source, so estimates are unbiased regardless of the pilot's accuracy. D) For contrast, a fixed rate spending a comparable overall budget ($k/\bar{r} = 0.20$) over-samples the dense catchment (32 sources) while drawing only 5 of the 14 nodes it should census.

a boundary street’s score from its neighbours’ traversals, and some of those neighbours sit just outside the boundary, so buffer streets must be eligible sources or boundary streets lose their contributions. *Betweenness* runs from every node in either mode, because a shortest path between two buffer nodes can credit a live node it passes through: closeness concerns only the study area’s own streets, whereas betweenness must also account for through movement. The test predicts each source’s traversal work from the size of the neighbourhood it explores, and selects sampling only when the predicted sampled work falls below a margin $\gamma = 0.75$ of the predicted exact work:

$$\sum_{u \in V} q_u \hat{r}_u^+ < \gamma \cdot \begin{cases} \sum_{u \in V_L} \hat{r}_u^+ & (\text{closeness only}) \\ \sum_{u \in V} \hat{r}_u^+ & (\text{betweenness requested}). \end{cases} \quad (5)$$

If exact computation costs the equivalent of 100 traversals, sampling engages only below a predicted cost of 75; a predicted cost of 80 would leave too little saving to cover the pilot’s overhead (Figure 2).

Work is priced from the *upper* confidence bound $\hat{r}_u^+$, a conservative high estimate of the population size, so a node the poll never hit is priced as the census it would receive rather than at zero. The work model omits fixed overheads such as the pilot; the margin γ covers them. The margin was set from preliminary timed runs on the four validation networks. It is the one empirically chosen constant besides the tolerance ε (Section 4.1), and it affects only whether sampling engages, not the estimates. In the validation, every cell the test approves for sampling runs faster than exact computation (Section 5).

Sampled closeness carries a built-in handicap: it must be able to draw sources from every node, live or buffer, whereas exact closeness runs only from the live nodes. On networks with a low live fraction, exact closeness is already cheap, so the work test routes closeness-only calls there to exact computation. Closeness sampling engages only on dense networks at long distances. Betweenness runs from all nodes in both modes, so its decision turns on the probabilities alone.

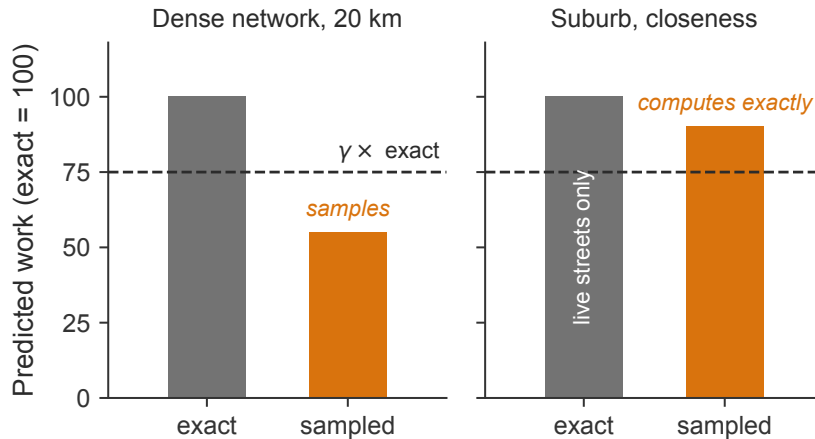


Figure 2: The work test compares predicted traversal work (illustrative values, exact = 100) and engages sampling only when the predicted saving clears the margin γ . On networks with a low live fraction exact closeness runs only the live streets and is already cheap, so the test declines to sample.

The test runs once per call and per distance. We call the resulting decision the *mode* of that computation: exact or sampled. A single-metric call carries its own mode; a call that computes both metrics together applies the with-betweenness pricing to both. The mode is thus a per-distance decision, and it is distinct from the per-node adaptivity of Section 3.2: within a sampled distance every node still carries its own inclusion probability, and nodes whose reach

the poll cannot certify as large saturate at $q = 1$, so sparse catchments are enumerated in full even when the distance as a whole is sampled.

3.4 Algorithm and worked example

Table 2 collects the parameters introduced in the preceding sections, with their defaults. Algorithm 1 restates the procedure for implementers using those defaults and introduces nothing beyond the preceding sections. Figure 1 illustrates the procedure on a worked example. In the implementation, distances routed to exact computation are batched, so a single Dijkstra per source serves all of them; each sampled distance runs its own pass under its own per-node probabilities.

Table 2: Parameters of the method. The tolerance ε is the one calibrated parameter (Section 4.1); the work-test margin γ is the one empirically chosen cost constant; the rest are fixed conventions of the implementation.

Symbol	Name	Role in the survey	Default
ε	Tolerance	How tight each street’s estimate must be	0.05
δ	Failure budget	The allowance for streets that miss the tolerance	0.1
α	Poll level	How cautious the reach bounds are	0.1
m	Poll size	How many pilot traversals run	$\min(n, \max(400, 0.025 n))$
$q_{\min}$	Probability floor	Keeps any one source from dominating	0.01
γ	Work-test margin	How large the predicted saving must be before sampling engages	0.75

3.5 What is guaranteed and what is validated

The design separates three levels of assurance: unbiasedness is guaranteed, additive accuracy for closeness is argued under stated approximations, and rank fidelity is demonstrated empirically. The Horvitz–Thompson weighting guarantees unbiasedness for both metrics, independently of pilot quality (Appendix A). The additive error guarantee of Section 3.1 applies to closeness and is stated for the idealised survey; the deployed design targets the same expected respondent count, and the gap between the two is assessed empirically. Rank fidelity, the quantity applications rely on, has no formal bridge from the additive bound and is validated in Section 5. Betweenness has no comparable per-source bound: a single source can account for a large share of a target’s value, so betweenness is judged entirely by the validation.

4 Ablation: why measurement is necessary

The load-bearing component of the method is the measurement of reach. This section tests that claim by ablation, replacing the measured reach with cheaper stand-ins. Three ways to know a population’s size form a ladder: assume it from a formula, count it on a map, or measure it on the network. Each of the first two rungs fails in a way the next rung catches; Table 3 maps the ladder.

4.1 Rung 1: assume reach from distance

A network-agnostic schedule can be built by assuming a single reach value per distance. On a regular grid with inter-node spacing s , the number of nodes within distance d is the area of a

Algorithm 1 Per-node reach-based sampling for localised network centrality

Require: Network $G = (V, E)$ with live nodes $V_L \subseteq V$ and buffer nodes $V_B = V \setminus V_L$

Require: Distance thresholds $D = \{d_1, \dots, d_j\}$

Require: Tolerance ε (default 0.05), failure probability δ (default 0.1), poll level α (default 0.1), work-test margin γ (default 0.75)

Ensure: Closeness estimates $\hat{C}_H[d]$ and betweenness estimates $\hat{C}_B[d]$ for each $d \in D$

```
1: Pilot: draw  $m = \min(|V|, \max(400, 0.025|V|))$  sources uniformly; run one Dijkstra per
   source, bounded at  $\max(D)$ ; record per-node hit counts  $h_u[d]$  for each  $d \in D$ 
2: for each distance  $d \in D$  do
3:   Reach bounds:  $\hat{r}_u^-, \hat{r}_u^+ \leftarrow$  one-sided Clopper–Pearson bounds from  $h_u[d]$  at level  $\alpha$ 
4:   Per-node probability:  $q_u \leftarrow 1$  if  $\hat{r}_u^- = 0$  (census); otherwise  $q_u \leftarrow$ 
    $\min(1, \ln(2\hat{r}_u^-/\delta) / (2\varepsilon^2 \hat{r}_u^-))$ , floored at 0.01
5:   Work test: exact cost  $\leftarrow \sum_{u \in V_L} \hat{r}_u^+$  (closeness only) or  $\sum_{u \in V} \hat{r}_u^+$  (with betweenness)
6:   if  $\sum_{u \in V} q_u \hat{r}_u^+ \geq \gamma \cdot \text{exact cost}$  then ▷ a census is cheaper; Section 3.3
7:     Run exact computation (closeness sources  $V_L$ ; betweenness sources  $V$ )
8:   else
9:     Sample sources  $S \subseteq V$  by including each node  $u$  independently with prob.  $q_u$ 
10:    for each source  $u \in S$  do ▷  $S$  includes both live and buffer nodes
11:      Run distance-bounded Dijkstra from  $u$ 
12:      Accumulate closeness to each live target  $v$  via  $1/(q_u \cdot d(u, v))$  ▷ IPW
13:      Run Brandes backpropagation; accumulate betweenness for live targets, weighted
         $1/q_u$ 
14:    end for
15:  end if
16: end for
17: return  $\{\hat{C}_H[d], \hat{C}_B[d] : d \in D\}$ 
```

Table 3: The ablation ladder: three ways to obtain reach.

Rung	Reach obtained by	Fails when	Shown in
1	Assumed from a grid formula	Real reach falls below it	Figure 3, Table 4
2	Counted on a map (Euclidean disc)	Circuitry crowds the disc rim	Figure 4
3	Measured on the network (pilot poll)	— (the method; Section 5)	Figure 5

circle divided by the area per node:

$$r_c = \frac{\pi d^2}{s^2}. \quad (6)$$

We call this the *canonical reach*, canonical in the sense of a fixed, network-agnostic reference assumption. With $s = 175$ m, Eq. 3 yields one fixed sampling rate per distance, $p(d) = \min(1, k(r_c)/r_c)$, shared by every node at that distance. The spacing is deliberately conservative: it sits above typical mean street-segment lengths in European cities (city means ≈ 95 – 122 m; Strano et al., 2013), and a larger s lowers the assumed reach and so raises the sampling rate, erring toward oversampling. The single free parameter, the tolerance ε , was calibrated once on the sparsest calibration network (Cary, NC), where the grid model is least conservative; $\varepsilon = 0.05$ is the loosest value at which Cary meets the $\rho \geq 0.95$ target at every distance, and the per-node method of Section 3 inherits it unchanged.

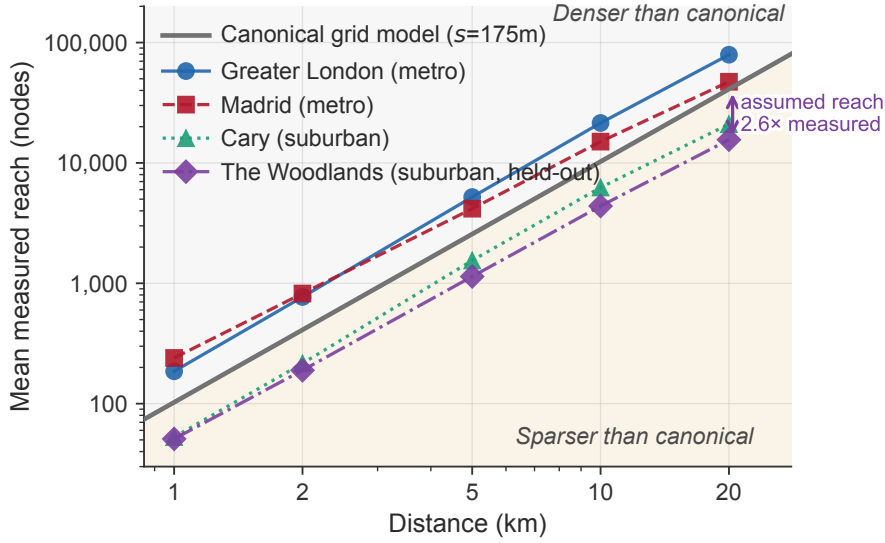


Figure 3: The grid assumption oversamples the metros and under-samples the suburbs, and the held-out network lies lowest of all, which predicts its failure (Table 4). Mean measured reach vs. analysis distance (logarithmic axes) for the four validation networks, against the canonical grid model ($s = 175$ m, grey line); background tints mark the regions above and below the canonical curve. The annotated arrow marks the held-out network’s 20 km gap, where the assumed reach is 2.6 times the measured value.

The assumption fails where real reach falls below the curve (Figure 3). Table 4 condenses the canonical schedule’s record; the per-distance detail is in Appendix Tables 8–11. The three calibration networks meet the target at every distance. The held-out network fails at the one distance where its closeness sampled: 20 km, with $\rho = 0.9386$ against the 0.95 target (shorter distances ran exact by construction). In the percentile-point terms of Section 2, passing cells hold the median street within 2.5 percentile points of its exact rank; on this cell the streets shift by 0.068 of the rank range on average, and the shifts cluster in space (Figure 10). The failure has a mechanistic explanation. At 20 km The Woodlands reaches 0.38 of the canonical reach, against 0.51 for Cary, so the schedule under-samples it more severely than any network used in calibration. Re-running the held-out network at a tighter tolerance restores the target, confirming under-sampling from overestimated reach rather than a defect of the bound. The schedule carries no information about an individual network’s reach; the measured reach identifies the deficit before any centrality run.

Table 4: Ablation rung 1 (assumed reach): the canonical schedule’s minimum Spearman ρ per network across 1–20 km. All minima occur at 20 km. The held-out network fails the $\rho \geq 0.95$ target for closeness; per-distance detail is in the appendix (Tables 8–11).

Network	Closeness (worst ρ)	Betweenness (worst ρ)
London	0.9927	0.9828
Madrid	0.9881	0.9817
Cary	0.9609	0.9654
Woodlands (held out)	0.9386 [†]	0.9576

[†]below the $\rho \geq 0.95$ target.

4.2 Rung 2: count reach on a map

A cheaper measurement counts, for each node, the neighbours within the Euclidean radius (a KD-tree query, no network traversal), then divides by a fixed deflation factor of 2.5, because Euclidean discs hold more nodes than network catchments of the same radius. The deflation exceeds the measured disc-to-reach ratio for all but a small tail of nodes (2.5–5.2% of live nodes per cell; `output/euclidean_reach_ratios.csv`). The consequence is asymmetric: over-deflation only oversamples, the cheap direction of error, while the tail nodes are under-sampled.

The count is correct on average but blind to circuitry and barriers alike. A node counts disc neighbours the network cannot deliver within the distance limit, whether a river severs them or ordinary route indirectness adds the excess travel. Figure 4 shows the mechanism at one Cary node: most of its disc population sits near the rim, beyond 10 km of network travel, so the true ratio exceeds the deflation and the node is under-sampled. The tail is too small to move a network-level rank correlation, but a map count cannot tell its members from typical nodes. The rung is therefore judged by the single-node mechanism rather than by an accuracy score.

4.3 Rung 3: measure reach on the network

The pilot poll of Section 3.2 travels on the network itself, so barriers, dead ends, and disconnected fringes are priced into the hit counts, and the confidence bounds convert the remaining uncertainty into oversampling rather than under-sampling. This rung is the method; there is nothing to ablate. Its results are the subject of Section 5.

5 Validation

5.1 Setup

We validate on four real-world street networks spanning the urban density range, each built from official open road-network data. *Greater London* (Ordnance Survey Open Roads; 156,434 live nodes) and *Greater Madrid* (the official *Red Viaria* road network; 64,610 live nodes) are dense metropolitan networks. *Cary, NC* (US Census TIGER/Line; 6,787 live nodes; mean node degree ≈ 2.5 , roughly the number of street segments meeting at a typical junction) is a low-density planned suburb in the Raleigh region. *The Woodlands, TX* (US Census TIGER/Line; 4,587 live nodes; mean node degree ≈ 2.5) is a master-planned suburb north of Houston with a strongly dendritic (tree-like, cul-de-sac-heavy) street pattern. Each network is analysed within a 20 km buffer (Section 2); in every case the live study area is delimited by the administrative boundary geocoded from OpenStreetMap via OSMnx (Boeing, 2017), independently of the road-geometry source.

The tolerance ε was calibrated on Cary, and the reach deflation of Section 4.2 on the calibration networks, so The Woodlands plays no part in selecting the accuracy parameters. It serves

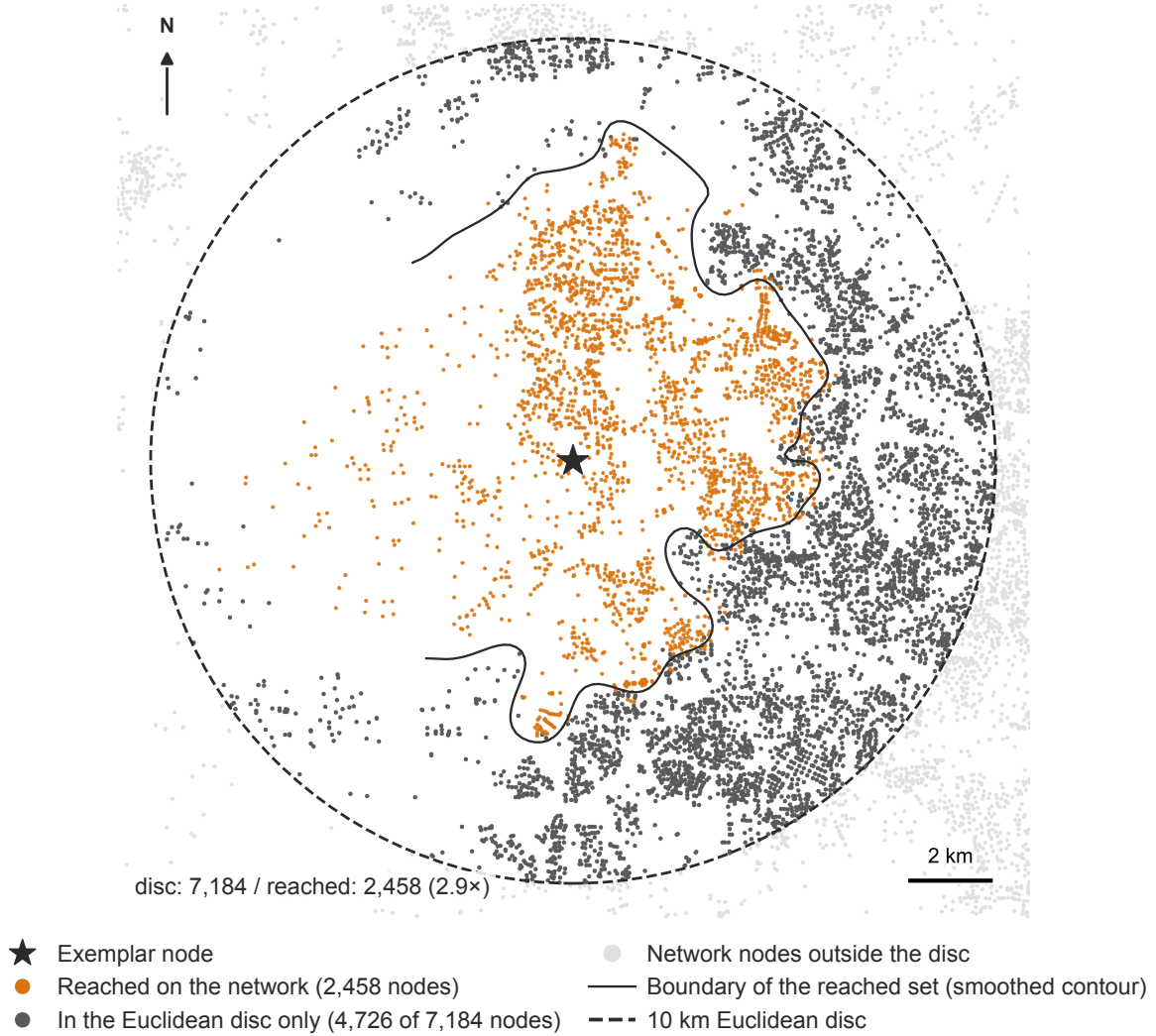


Figure 4: Ablation rung 2: a map count claims far more neighbours than the network can deliver, so the fixed deflation under-corrects and the node is under-sampled. Euclidean disc against network reach at 10 km for the Cary node with the largest disc-to-reach ratio: the disc (dashed circle) contains 7,184 nodes; the network traversal from the node reaches 2,458 (orange), with the solid line marking the smoothed boundary of the reached set. Grey nodes lie inside the disc but beyond 10 km of network travel. No single barrier severs the eastern block: the exclusion is ordinary path circuitry accumulating over a disc population concentrated near the rim. The 2.5 deflation absorbs typical circuitry; here the ratio rises to 2.9. The palest points are network nodes outside the disc, shown for context.

as the out-of-sample accuracy test. The one constant set from timings on all four networks is the work-test margin γ , which affects which distances sample but not the values of sampled estimates (Section 3.3); the held-out 20 km betweenness cell realises a cost far enough below exact that its sampling decision is insensitive to the margin’s value.

Ground truth uses all sources ($p = 1$) and is deterministic. All timings are wall-clock on one workstation (Apple M5 Pro, 18 cores, 48 GB memory); exact and sampled runs use the same parallel traversal code. Each speedup divides the exact reference time, recorded when the ground truth was built, by the mean sampled runtime. The two were timed in separate sessions on the same machine, and small timing differences are sensitive to background load, so the smallest ratios (the 5 km betweenness cells) are the least certain; the mode decisions and the headline 10–20 km speedups do not depend on them.

The validation runs each metric at each distance as a separate single-metric call, so the work test’s mode is recorded per metric and every sampled cell pays a full pilot; a multi-distance call shares one pilot and costs less. We report the mean over three random seeds; the largest inter-seed standard deviation in ρ across all networks, metrics, and distances is 0.0019.

5.2 Accuracy and cost

All four networks meet the $\rho \geq 0.95$ target for both metrics at every distance from 1–20 km (Figure 5; Table 5 consolidates ρ and measured wall-clock speedups). The minimum sampled value is $\rho = 0.9619$, on the held-out network’s 20 km betweenness; distances the work test routed to exact computation equal the ground truth by construction (open markers in Figure 5). The inter-seed spreads (at most 0.0019) are small enough that every individual seed also clears the target, and the held-out network passes everywhere with no additional tuning. Panel C validates that network on a 50 km buffer across the full 1–50 km range; Section 6.3 reports the extended distances.

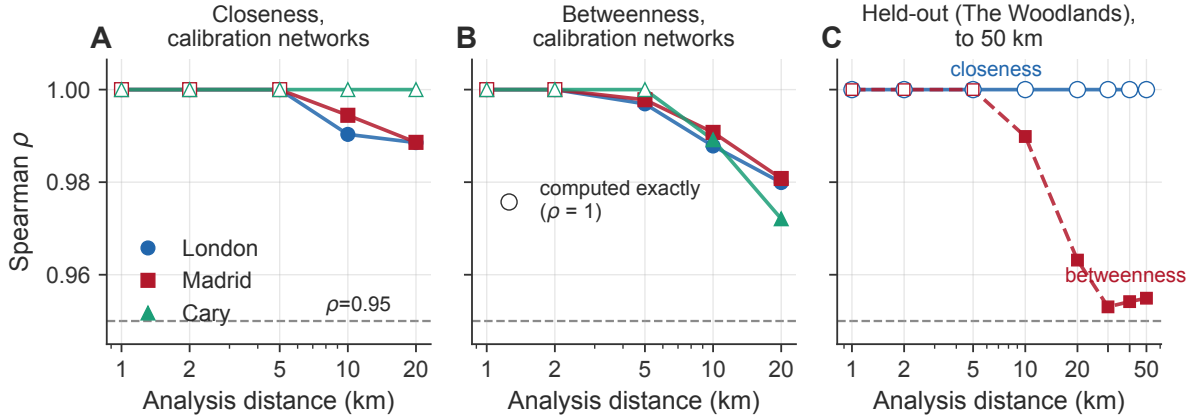


Figure 5: Every point on every network meets the $\rho \geq 0.95$ target (dashed line), the held-out network included. Spearman ρ vs. analysis distance under the per-node method. Panel C validates the held-out network across the full 1–50 km range on a 50 km buffer, which supports the thresholds beyond the 20 km extent of the main validation. Open markers denote distances the work test routed to exact computation, where $\rho = 1$ by construction; filled markers are sampled.

The work test’s decisions follow distance and density (per-cell modes in Table 5). No network samples at 1–2 km, where traversals are cheap; the metropolitan networks sample betweenness from 5 km and closeness from 10 km; the suburbs sample only betweenness, at 10–20 km, because their low live fractions make exact closeness cheap (Section 3.3). The suburbs’ exact closeness entries record the work test operating as designed: exact closeness on those networks is already

Table 5: Per-node method on the four validation networks. Entries marked exact record the work test declining to sample by design, where exact computation is already cheaper (Section 3.3); they equal the ground truth, so no timing is reported for them. All speedups include the pilot’s cost. ($\varepsilon = 0.05$, $\delta = 0.1$; mean of three seeds; each metric ran through its single-metric entry point, so the decision is recorded per metric. Speedups are wall-clock ratios of the exact runtime to the sampled runtime.)

Network	Dist.	ρ_c	Speed-up _c	ρ_b	Speed-up _b
London	1 km	exact	—	exact	—
London	2 km	exact	—	exact	—
London	5 km	exact	—	0.9969	1.3×
London	10 km	0.9903	2.9×	0.9878	4.2×
London	20 km	0.9885	11.8×	0.9799	15.5×
Madrid	1 km	exact	—	exact	—
Madrid	2 km	exact	—	exact	—
Madrid	5 km	exact	—	0.9978	1.2×
Madrid	10 km	0.9945	1.6×	0.9908	2.6×
Madrid	20 km	0.9886	5.7×	0.9808	9.2×
Cary	1 km	exact	—	exact	—
Cary	2 km	exact	—	exact	—
Cary	5 km	exact	—	exact	—
Cary	10 km	exact	—	0.9892	1.7×
Cary	20 km	exact	—	0.9721	5.6×
Woodlands	1 km	exact	—	exact	—
Woodlands	2 km	exact	—	exact	—
Woodlands	5 km	exact	—	exact	—
Woodlands	10 km	exact	—	0.9899	4.3×
Woodlands	20 km	exact	—	0.9619	14.2×

inexpensive, and sampling would add cost.

Every sampled cell runs faster than exact computation, and the gain grows with distance (Table 5): the smallest speedup is Madrid’s 5 km betweenness, where the pilot absorbs most of the predicted saving, and by 20 km the sampled cells run $5.6\times$ to $15.5\times$.

On the cells both designs sample, the comparison with the canonical schedule turns on how a network’s measured reach compares with the schedule’s assumed reach (per-cell numbers in Tables 8–11). London’s measured reach exceeds the canonical assumption (Figure 3), so the per-node probabilities sit below the schedule’s fixed rate and the method is faster at 20 km. On Madrid the two designs run close together. On suburban betweenness the schedule is faster: its rate derives from an assumed reach above the suburbs’ measured reach, so it draws fewer sources, and it pays no pilot and no separate pass per sampled distance. The same under-sampling fails the held-out closeness target (Section 4.1), and the schedule cannot detect the failure in advance.

Figure 6 places the method next to the canonical schedule of Section 4.1: the held-out closeness failure disappears, and the suburb closeness entries are computed exactly. Where both designs look worse (panel C), the streets involved carry almost no traffic and their relative order is meaningless: many low-reach streets have betweenness values at or near zero, so the reach-quartile decomposition shows a within-quartile depression shared by both designs at any sample size.

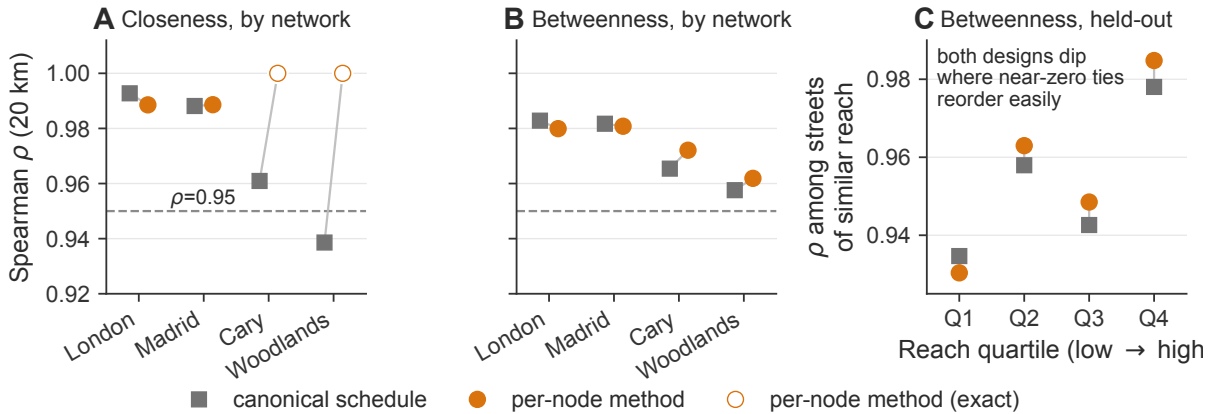


Figure 6: Canonical schedule (ablation rung 1) versus the per-node method at 20 km. In panels A and B the rung-1 held-out failure (Woodlands closeness, below the target line) is resolved; open markers denote per-node closeness entries the work test routed to exact computation; values are means over three seeds (Table 5). Where both designs score lower in panel C, the streets involved carry almost no traffic and their relative order is meaningless: the within-quartile depression comes from tie-heavy near-zero values rather than a sample deficit, and both designs share it. The upper three quartiles improve slightly and the lowest-reach quartile shifts by a similar small amount the other way. Panel C omits the target line because the target applies to the full ranking rather than to within-quartile correlations.

5.3 Error structure

The surviving error has little effect on rankings; the appendix details its structure (Figures 7–10). Median relative error is roughly constant across reach deciles and lowest in the top decile (Figure 8), so the highest-value nodes are estimated with the best relative precision, and their ranks barely move because centrality spans orders of magnitude. Betweenness relative error is nevertheless large in the middle deciles, reaching tens of percent on the suburban networks: the true values there sit at or near zero, so small absolute deviations register as large relative errors while barely moving the ranking, and the unbiased weighting makes this error noise around the

truth rather than drift. The rank statistics in the same cells stay tight. Rank shifts are small and fall mostly on low-centrality nodes (Figure 7), and most nodes stay in their exact-computation decile (Figure 9). Correlations computed within reach quartiles sit below the network-wide values, largely through range restriction (values within a quartile are close together, so small errors reorder them easily); the within-quartile rank shifts themselves stay small (Appendix, error structure detail). On the held-out network at 20 km, the canonical schedule’s closeness errors cluster in space, whereas the method computes that cell exactly and leaves no residuals (Figure 10). Its betweenness errors are similar in size under the two designs but cluster less under the method.

6 Discussion

6.1 Practical guidance

In `cityseer`, sampling exposes a single configurable parameter, the tolerance ε , set by default to $\varepsilon = 0.05$; the library polls the network, assigns per-node probabilities, and selects exact computation where sampling cannot pay, with no further input. The default requires no adjustment, and a user who wants a different balance of speed and precision can change ε as described below. Algorithm 1 gives the full procedure for other implementations.

The realised speedup depends on distance, the live fraction, and network size. Distance matters most, because gains concentrate at long thresholds (Table 5). The live fraction governs closeness: where few of a network’s nodes are live, exact closeness is already cheap, and the work test routes it to exact computation. Network size sets a floor: below about 1,000 nodes exact computation completes in seconds and sampling is unnecessary.

Precision and speed trade off through the tolerance ε : halving it quadruples the required samples, and a looser $\varepsilon = 0.08$ – 0.1 buys speed for exploratory work at some cost to ranking accuracy. Betweenness bounds how loose ε can go: it has heavier-tailed contributions (a single origin can dominate a street’s betweenness) and no per-source bound, so it is the first metric to degrade. On an unfamiliar network type, a check against exact computation at one long distance confirms $\rho > 0.95$ before a full analysis commits to the sampled path.

6.2 What the guarantee does and does not cover

The Hoeffding bound controls absolute error for the idealised survey of Section 3.1; the validation measures whether the *ranking* of nodes is preserved, and no formal bound connects the two. The bound is loose in practice, for three reasons. Hoeffding’s inequality uses only the range of contributions, not their much smaller variance, and the tolerance is stated relative to a catchment-level bound that can be large in absolute terms (Appendix A). The failure-budget allocation treats node-level errors as independent, though they are positively correlated. And rank preservation is more forgiving than additive error, because centrality values span several orders of magnitude (Section 5.3). The pilot’s confidence bounds push the remaining uncertainty toward oversampling, and the default ε was chosen by rank-fidelity calibration on Cary (Section 4.1). The pair (ε, δ) therefore describes a conservative operating point rather than a certified additive error.

6.3 Scope and limitations

The method applies to shortest-path (metric) distance. Angular (simplest-path) distance (Turner, 2007) does not satisfy the subpath optimality assumed in the betweenness derivation and remains to be tested. Validation spans dense European metropolitan networks and two low-density US suburbs; morphologies outside this range (informal settlements, gridiron megacities) are untested.

The main four-network validation ends at 20 km, the largest threshold its buffered extents support. To test beyond it, the held-out network is rebuilt with a 50 km buffer and validated across the full 1–50 km range (Figure 5C): the work test routes closeness to exact computation throughout, and sampled betweenness holds $\rho = 0.953\text{--}0.955$ through 30, 40, and 50 km at speedups of 6–14 $\times$ (`output/woodlands_frontier.csv`). The decline in ρ with distance flattens just above the target rather than crossing it.

Rank-based evaluation has no formal bridge from the additive bound, and betweenness has no per-source bound; both are validated rather than proven. The sampled closeness evidence comes from the metropolitan networks, because on the suburbs the work test routes closeness-only calls to exact computation at every distance; the held-out sampled evidence is therefore betweenness. A joint call can nevertheless sample suburban closeness (Section 3.3), and a supplementary check covers that configuration: forcing the sampled closeness path past the work test at 20 km meets the target on both suburbs, with every seed above it (three seeds each; `output/forced_closeness.csv`).

Two residual effects survive the design. The first is the pilot’s tail risk: a node the poll happens to hit unusually often has its catchment overestimated and is under-sampled, with probability bounded by α per node; the other error directions are safe. Because every node is estimated from the same pilot draw, these errors are correlated: an unlucky draw overestimates many nodes at once, though by small margins (Appendix, pilot coverage). The validation seeds include such draws. The second is the betweenness quartile gradient of Section 5, which improves under the method but does not vanish. A diagnostic on the held-out network at 20 km traces it to the high-traffic streets: nodes crossed by 100 or more of the 400 pilot traversals shift by a median of 1.7 and a 90th percentile of 9.6 percentile points, while the low-traffic periphery holds stable. The surviving noise is therefore the ordering of high-traffic streets, where a few origins dominate an arterial’s value.

Natural extensions include a concentration-aware pilot for betweenness, validation on angular distance and a broader range of morphologies, online stopping criteria (Borassi and Natale, 2019), and variance-aware bounds (Pellegrina and Vandin, 2023).

6.4 Related work

Table 6 summarises existing sampling approaches, grouped by computational primitive; Matta et al. (2019) provide a comprehensive empirical comparison for global betweenness. Prior methods differ from ours on two axes: they set one network-wide precision target rather than a per-catchment one, and most estimate a single metric rather than sharing one source sample across both. *Path-sampling* applies only to betweenness, each sample updates only one path, and its adaptive variants (Riondato and Upfal, 2018; Borassi and Natale, 2019) adapt to a global additive target, as does the k -bounded-distance variant of Riondato and Kornaropoulos (2016); only *source-sampling* updates every reached node from a single traversal, so only it can serve both metrics from one sample sized to each catchment. All-distances sketches (Cohen et al., 2014; Cohen, 2015) are the closest prior for the closeness half: they give per-node relative-error guarantees and accept distance cutoffs, but they serve no betweenness, estimate every node rather than selecting exact computation where a census is cheaper, and require a dedicated sketch construction rather than the bounded Dijkstra traversals of standard centrality tooling.

7 Conclusion

Localised centrality is estimated over each node’s own catchment, so precision must follow the catchment. The method presented here measures every catchment with a pilot poll, sizes each node’s inclusion probability from a confidence bound on its reach, keeps estimates unbiased with inverse-probability weighting, and selects exact computation wherever a census is cheaper

Table 6: Sampling approaches for centrality approximation. *Source-sampling* traverses the network from each sampled node, updating all reached nodes. *Path-sampling* draws individual shortest paths, updating only nodes along each path. Prior guarantees are uniform over the network; our approach uses distance-bounded reach r instead of the full node count n and serves both metrics from a single sample.

Method	Metric	Guarantee	Samples k	Key param.
<i>Source-sampling (full traversal per sample)</i>				
Eppstein & Wang (2004)	Closeness	Additive, global	$O(\log n/\varepsilon^2)$	n
Brandes & Pich (2007)	Both	Empirical	User-chosen	—
Bader et al. (2007)	Betw.	Relative, per query	$O(1/(\varepsilon^2 b(v)))$	$b(v)$
Geisberger et al. (2008)	Betw.	Heuristic (dist.-scaled)	User-chosen	—
<i>Path-sampling (single shortest path per sample)</i>				
Riondato & Kornaropoulos (2016)	Betw.	Additive, global	$O(\log VD/\varepsilon^2)$	VD
ABRA (2018)	Betw.	Additive, global	Adaptive (Rademacher)	—
KADABRA (2019)	Betw.	Additive, global	Adaptive	D
SILVAN (2023)	Betw.	Non-uniform	Variance-dep.	Variance
This work	Closeness	Additive, local*	$O(\log r/\varepsilon^2)$	r
(this work)	Betw.	Unbiased; rank-validated	(shared sample)	r

*Union bound over the r nodes within each distance-bounded catchment, not over all n nodes globally (Section 3.1). Betweenness shares the closeness sample; it is unbiased under Horvitz–Thompson weighting, has no per-source bound, and is judged empirically (Section 3.5).

than a survey, all governed by one tolerance parameter. On four street networks spanning the urban density range, with one held out from accuracy calibration, the method preserves rank ($\rho \geq 0.96$) for closeness and betweenness at 1–20 km. The ablation shows the cheaper stand-ins for measured reach failing exactly where the reach measurements predict. The remaining caveats are the rank-based evaluation, the absence of a per-source bound for betweenness, and the untested angular case.

The method is implemented in the open-source **cityseer** Python package.¹ Analysis scripts and validation data are available in the repository’s **analysis/sampling/** directory.

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¹github.com/benchmark-urbanism/cityseer-api

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A Technical Details

GLA network preprocessing. The Greater London Authority road network is sourced from Ordnance Survey Open Roads data (GeoPackage format).

Preprocessing steps:

1. Remove invalid and empty geometries
2. Explode multi-part geometries
3. Build NetworkX graph from edges
4. Remove filler nodes (degree-2 nodes that do not represent junctions)
5. Remove dangling nodes (dead-end segments < 20 m)
6. Consolidate nearby nodes (threshold 1 m)
7. Extract largest connected component

Final network: 311,392 nodes, of which 156,434 are live within the boundary.

Madrid network. The Greater Madrid metropolitan network (64,610 live nodes) is sourced from the official *Red Viaria* road-network dataset for the Community of Madrid and preprocessed using the same consolidation pipeline as the GLA network. As for the other networks, the boundary delimiting the live area is geocoded from OpenStreetMap via OSMnx (Boeing, 2017).

Cary and Woodlands networks. The Cary, NC network (6,787 live nodes) and The Woodlands, TX network (4,587 live nodes) are built from US Census TIGER/Line *edges*, the topologically integrated road layer, which is natively noded at intersections (the cartographic *roads* layer is not, and does not build a connected graph). Road edges (`ROADFLG = Y`) are loaded for every county intersecting the analysis area plus its 20 km buffer, then passed through the same consolidation pipeline as the GLA network. The live area is the municipal boundary; the surrounding region supplies the buffer.

Pilot reach estimator and confidence bounds. On an undirected network the probability that a uniformly drawn pilot source lies within distance d of node u equals $r_u(d)/n$, so u ’s hit count over the m pilot traversals is a binomial-type estimator of its reach at every distance. The lower and upper reach bounds $\hat{r}_u^-$ and $\hat{r}_u^+$ are one-sided exact binomial (Clopper–Pearson) bounds at level $\alpha = 0.1$ (Clopper and Pearson, 1934). The pilot draws without replacement, so its hit counts sit slightly more tightly around the truth than independent draws would, and the binomial bounds stay on the conservative side (Hoeffding, 1963, Theorem 4).

A direct check compares the bounds against the exact reach on all four networks (ten pilot draws each; `pilot_coverage_check.py`). Table 7 summarises the 20 km cells. The point estimate is unbiased and accurate, with a median relative error of 1.5–3% at 20 km. Mean lower-bound coverage sits near the nominal 0.90. Because every node is estimated from the same draw, coverage is correlated across nodes: a single unlucky draw overestimates many nodes at once, so single-draw coverage ranges widely. The overshoot in such draws is small, a median of 1–3% beyond 5 km, which perturbs the resulting inclusion probabilities by a similar margin; the end-to-end validation, whose seeds include such draws, absorbs it.

Table 7: Pilot reach-bound coverage at 20 km, ten pilot draws per network against the exact reach. Coverage is the share of live nodes whose lower confidence bound sits at or below the true reach (nominal 0.90 at $\alpha = 0.1$). All nodes share each draw, so coverage is correlated across nodes and single draws range widely; violations overshoot the true reach by the stated median ratio. Per-distance detail in `output/pilot_coverage.csv`.

Network	Coverage (mean)	Single-draw range	Violation ratio	Point median rel. err.
London	0.86	0.68–1.00	1.006	0.015
Madrid	0.93	0.81–1.00	1.015	0.018
Cary	0.91	0.31–1.00	1.011	0.026
The Woodlands	0.84	0.17–1.00	1.011	0.032

Inverse-probability weighting. When nodes are sampled as sources with per-node inclusion probabilities q_u , each sampled contribution is scaled by the reciprocal of its own inclusion probability. This is a standard Horvitz–Thompson estimator (Horvitz and Thompson, 1952) and is unbiased whenever the probabilities are positive; unbiasedness does not depend on how the probabilities were chosen, so a rough pilot reach estimate affects only variance. In the idealised argument of Section 3.1, $C_{\max} = r_v/d_{\text{near}}$ suffices for harmonic closeness, where d_{near} is the smallest distance from v to a catchment member; each weighted term then lies in $[0, C_{\max}/k]$, and Hoeffding’s inequality covers these without-replacement draws. $C_{\max}$ can be large where a neighbour sits close to v , so a tolerance stated relative to $C_{\max}$ can be loose in absolute terms (Section 6.2). In the deployed design the probability floor ($q_u \geq 0.01$) bounds the weights at 100, so a source u ’s weighted term for target v is at most $100/d(v, u) \leq 100/d_{\text{near}}$, the deployed-design analogue of the per-term bound. The source pool includes both live and buffer nodes (Section 3.3), ensuring that every node which contributes in exact mode has a positive probability of being sampled.

For closeness, sources enter the sample by independent per-node draws, and each source’s contribution to a target is a fixed quantity (the inverse distance), so the per-source terms of the estimator are independent. For betweenness, the Brandes decomposition yields a fixed per-source term in the same way, so the same argument applies.

Betweenness over all routes. Betweenness counts shortest paths from every node, live or buffer: a path between two buffer nodes that passes through the interior legitimately credits the live intermediate it traverses, so excluding it would under-count boundary nodes. The *live* designation therefore filters only which nodes’ values are reported, not which sources run. Each ordered pair (s, t) contributes with weight $\frac{1}{2}$ on undirected graphs (the two orderings of a symmetric pair are counted once between them) and 1 on directed graphs; buffer-to-buffer routes through the interior are included. Under sampling this estimand is unchanged, and the Horvitz–Thompson $1/q_u$ scaling applies uniformly with no per-pair correction.

Implementation in cityseer. The method is implemented in the `cityseer` Python package (version 4.25+) and enabled via the `sample=True` flag. For each call, the library polls the network

once (`estimate_polled_reach`), assigns per-node probabilities per distance (`compute_node_p`), and applies the work test of Section 3.3 once per call and per distance. Distances routed to exact computation are batched into a single pass; each sampled distance runs its own pass under its own per-node probabilities. The `centrality_shortest` function computes both closeness and betweenness from shared Dijkstra traversals in a single call:

```
from cityseer.metrics import networks

gdf = networks.centrality_shortest(
    network_structure,
    nodes_gdf,
    distances=[500,1000,2000,5000,10000,20000],
    sample=True
)
```

Convenience wrappers (`closeness_shortest`, `betweenness_shortest`) compute a single metric each; the validation of Section 5 ran these wrappers, so they are the tested configuration. A joint `centrality_shortest` call shares its traversals between the metrics and costs less; its mode decision can sample suburban closeness, the configuration checked by the forced-sampling run of Section 6.3.

The defaults ($\varepsilon = 0.05$, $\delta = 0.1$, poll level $\alpha = 0.1$, poll size $m = \min(n, \max(400, 0.025n))$, probability floor 0.01) are defined in `cityseer.sampling`, which also provides helper functions for custom workflows:

- `estimate_polled_reach`: the pilot poll, returning lower-bound, point, and upper-bound reach estimates per distance
- `compute_node_p`: per-node inclusion probabilities from reach estimates
- `compute_hoeffding_p`: the reach-to-probability bound (Eq. 3)
- `compute_distance_p`: the canonical grid reference model of Section 4.1, retained for the ablation

Reproducibility. All analysis scripts and validation results are available at `analysis/sampling/`. The pipeline runs via `run_all.py`: `01.validate_gla.py` through `04.validate_woodlands.py` build the exact ground truths and the canonical-schedule ablation record (the schedule is driven through the low-level API with an explicit per-distance probability, since the library’s `sample=True` flag now runs the per-node method), `validate_adaptive.py` runs the per-node method, and 05–10 generate the figures, macros, and tables. Supplementary checks run separately: `measure_disc_reach_ratio.py` (Section 4.2), `forced_closeness_check.py` (Section 6.3), and `epsilon_sweep.py`. Each script’s outputs are documented in its header.

Canonical schedule: per-distance ablation record. Tables 8–11 record the canonical schedule of Section 4.1 on each network. Rows marked “exact” have $p \geq \varphi$, the schedule’s fallback threshold; the rule is sized to exact closeness cost and applies to the whole call, so both metrics run exact in those rows and the schedule is not exercised. The held-out failure discussed in Section 4.1 is the 20 km closeness row of Table 11.

Error structure detail. The per-node rank-shift maps (Figure 7), the error-vs-reach deciles (Figure 8), the decile-transition matrices (Figure 9), and the paired rank-shift maps (Figure 10) support the error-structure summary of Section 5.3. The supporting statistics are as follows. Across the sampled metropolitan closeness cells the minimum within-quartile rank correlation

Table 8: Sampling validation on Greater London network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m, $\varphi = 0.50$).

Dist.	p	ρ_c	Spd_c	ρ_b	Spd_b
1 km	exact	1.0000	—	1.0000	—
2 km	exact	1.0000	—	1.0000	—
5 km	exact	1.0000	—	1.0000	—
10 km	23.8%	0.9944	2.0×	0.9925	3.1×
20 km	6.6%	0.9927	8.2×	0.9828	11.2×

Network: Greater London network, 156,434 nodes. Rows marked “exact” have $p \geq \varphi$, the schedule’s fallback threshold; the rule is sized to exact closeness cost and applies to the whole call, so both metrics run exact in those rows. Subscripts: c = closeness, b = betweenness. Notes apply to Tables 8–11.

Table 9: Sampling validation on Greater Madrid network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m, $\varphi = 0.36$).

Dist.	p	ρ_c	Spd_c	ρ_b	Spd_b
1 km	exact	1.0000	—	1.0000	—
2 km	exact	1.0000	—	1.0000	—
5 km	exact	1.0000	—	1.0000	—
10 km	23.8%	0.9945	1.5×	0.9934	3.1×
20 km	6.6%	0.9881	6.0×	0.9817	10.1×

Network: Greater Madrid network, 64,610 nodes.

Table 10: Sampling validation on Cary, NC (suburban) network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m, $\varphi = 0.12$).

Dist.	p	ρ_c	Spd_c	ρ_b	Spd_b
1 km	exact	1.0000	—	1.0000	—
2 km	exact	1.0000	—	1.0000	—
5 km	exact	1.0000	—	1.0000	—
10 km	exact	1.0000	—	1.0000	—
20 km	6.6%	0.9609	1.6×	0.9654	9.7×

Network: Cary, NC (suburban) network, 6,787 nodes.

Table 11: Sampling validation on The Woodlands, TX (held-out suburban) network ($\varepsilon = 0.05$, $\delta = 0.1$, $s = 175$ m, $\varphi = 0.11$).

Dist.	p	ρ_c	Spd_c	ρ_b	Spd_b
1 km	exact	1.0000	—	1.0000	—
2 km	exact	1.0000	—	1.0000	—
5 km	exact	1.0000	—	1.0000	—
10 km	exact	1.0000	—	1.0000	—
20 km	6.6%	0.9386	2.4×	0.9576	16.2×

Network: The Woodlands, TX (held-out suburban) network, 4,587 nodes.

is 0.876 (per-quartile detail in the validation CSVs), and the largest within-quartile median rank shift at 20 km is 2.9 percentile points; the rank-shift statistic does not depend on value separation, so it is the fairer within-quartile measure. On the held-out network at 20 km under the canonical schedule, the correlation between a node's closeness rank shift and the mean shift of its eight nearest neighbours is 0.55 and the mean rank displacement is 0.068 of the rank range; the per-node method computes that cell exactly. For betweenness the mean rank displacement is 0.048 under the schedule and 0.051 under the method, while the neighbour correlation falls from 0.25 to 0.18.

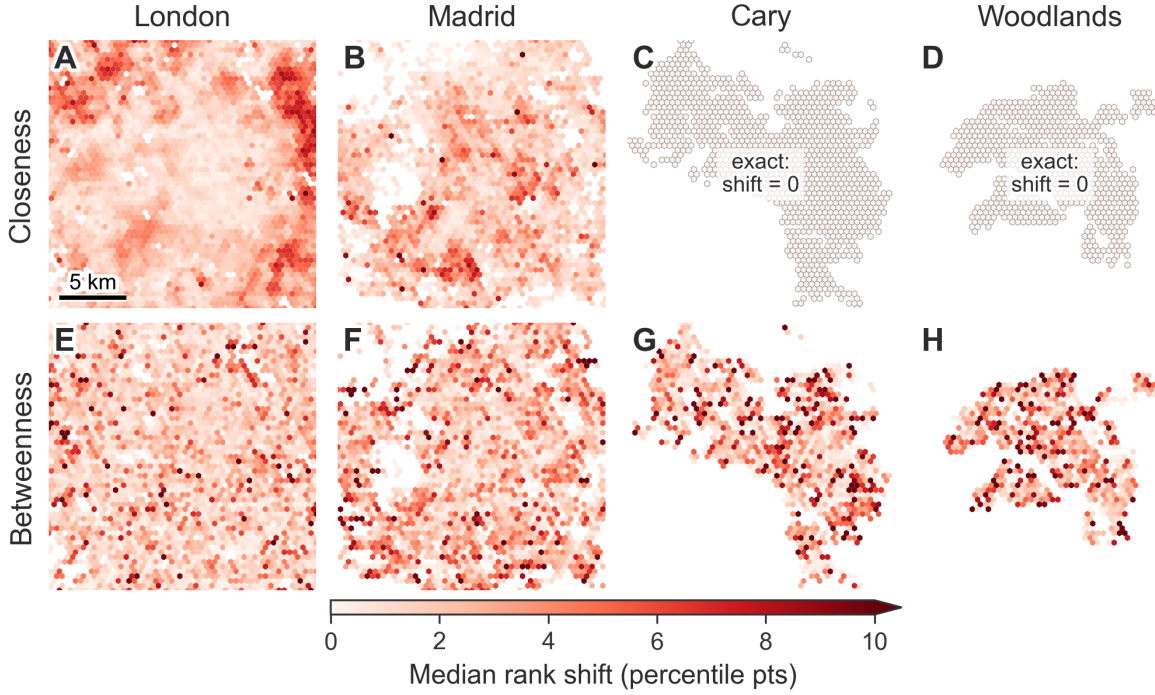


Figure 7: Per-node *rank shift* under the per-node method at 20 km: $|\text{pctile}(\text{true}) - \text{pctile}(\text{sampled})|$ in percentile points, drawn as hexbin medians on a fixed 0–10 scale; cells above 10 saturate (largest medians: 14.5 points, Madrid closeness; 33.4, Cary betweenness; saturated cells at most 0.2% of each closeness panel and 1.1–8.4% of each betweenness panel). Each panel shows a 20 km window centred on the network, drawn at a common scale, so one scale bar serves all panels. Shifts are larger for betweenness and fall mostly on low-centrality nodes, whose tied values flip easily (Figure 9); panels C and D show zero shift because the suburb closeness cells ran exactly.

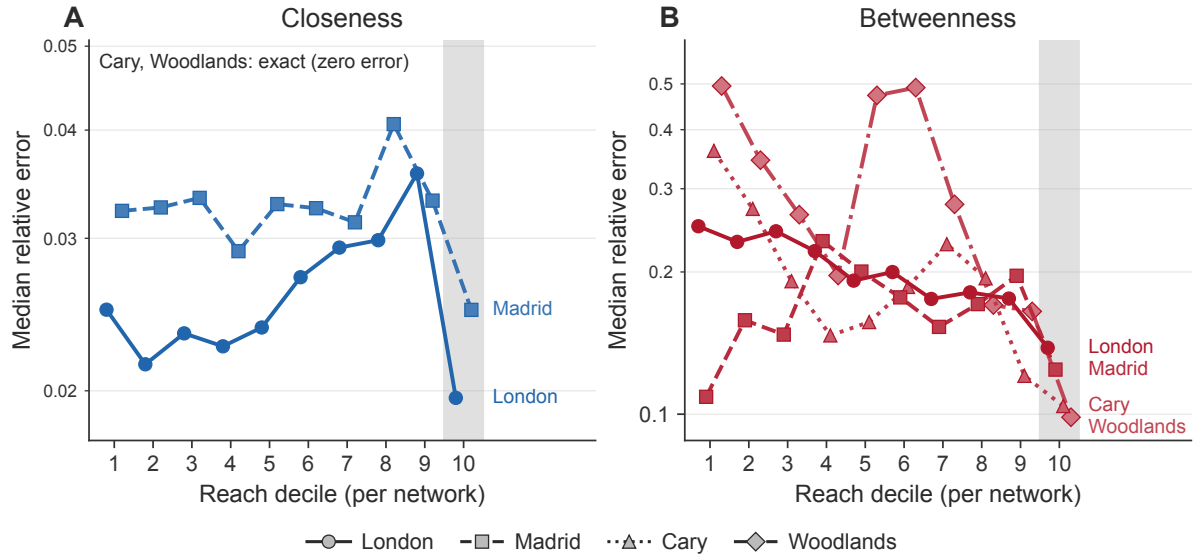


Figure 8: Median per-node relative error ($|\text{estimate} - \text{true}| / \text{true}$, zero-true nodes excluded) by reach decile under the per-node method at 20 km, for closeness (A) and betweenness (B). Each network is binned by its own reach distribution, so a given decile spans different reach values on different networks. Betweenness relative error runs roughly an order of magnitude above closeness error (note the differing y-axes). Panel A shows the metropolitan networks only: the suburban closeness cells ran exactly, leaving zero error and nothing to draw on a log axis. Median relative error is roughly constant across deciles and lowest in the top decile, marked by the shaded vertical band, so the highest-value nodes are estimated with the best relative precision. The held-out network's betweenness curve bumps upward in the middle deciles; its sparse mid-deciles hold few nodes, so the median there is noisy.

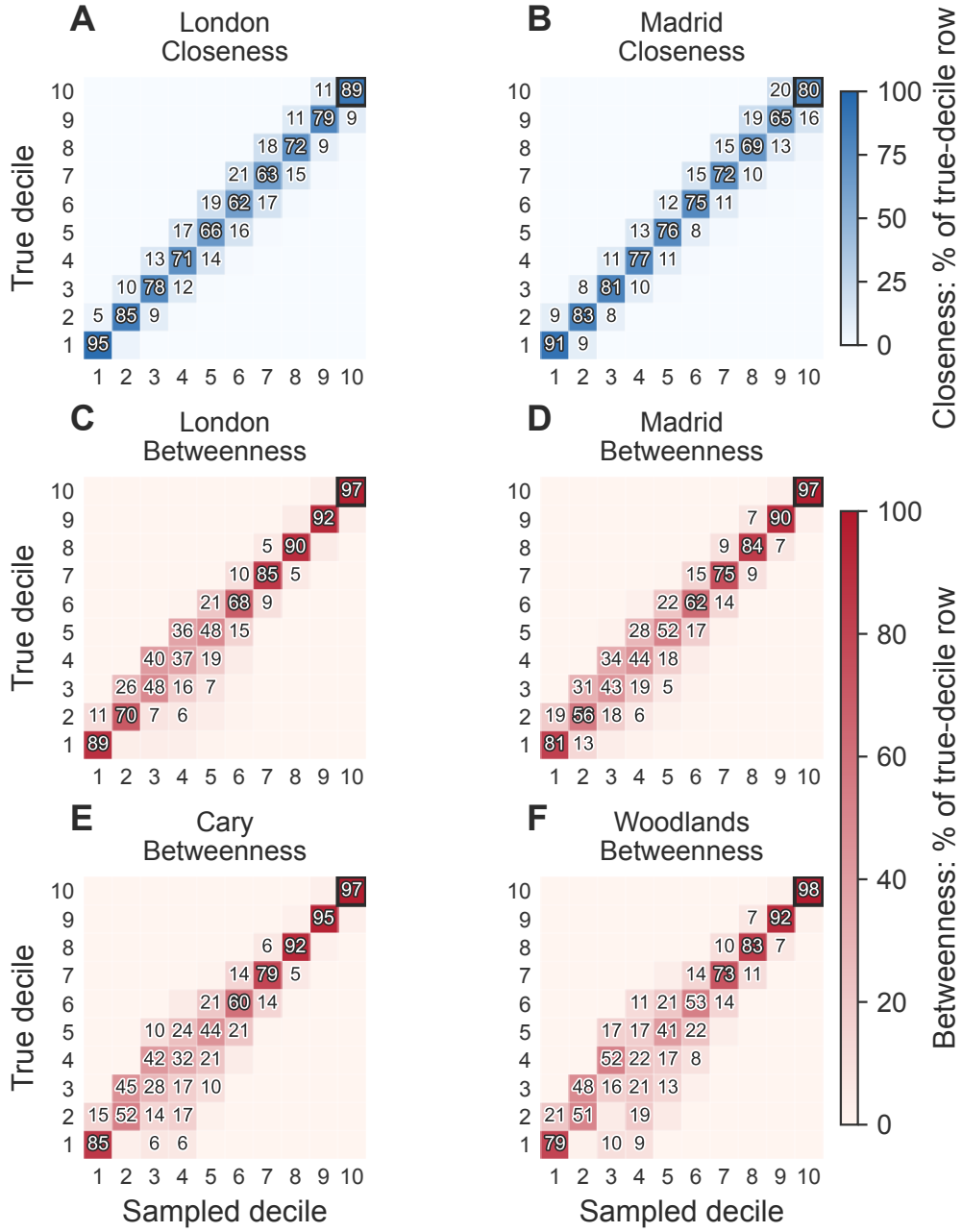


Figure 9: Decile transition matrices under the per-node method at 20 km: the proportion of nodes remaining in their true centrality decile (diagonal) or moving to another; each row sums to 100%. Panels show metropolitan closeness and betweenness (A–D) and suburb betweenness (E, F); the suburb closeness cells ran exactly and are omitted. Diagonal dominance is strongest in the upper deciles; the tie-heavy lower-middle betweenness deciles hold near-zero values that re-order easily, and top-decile retention stays high throughout.

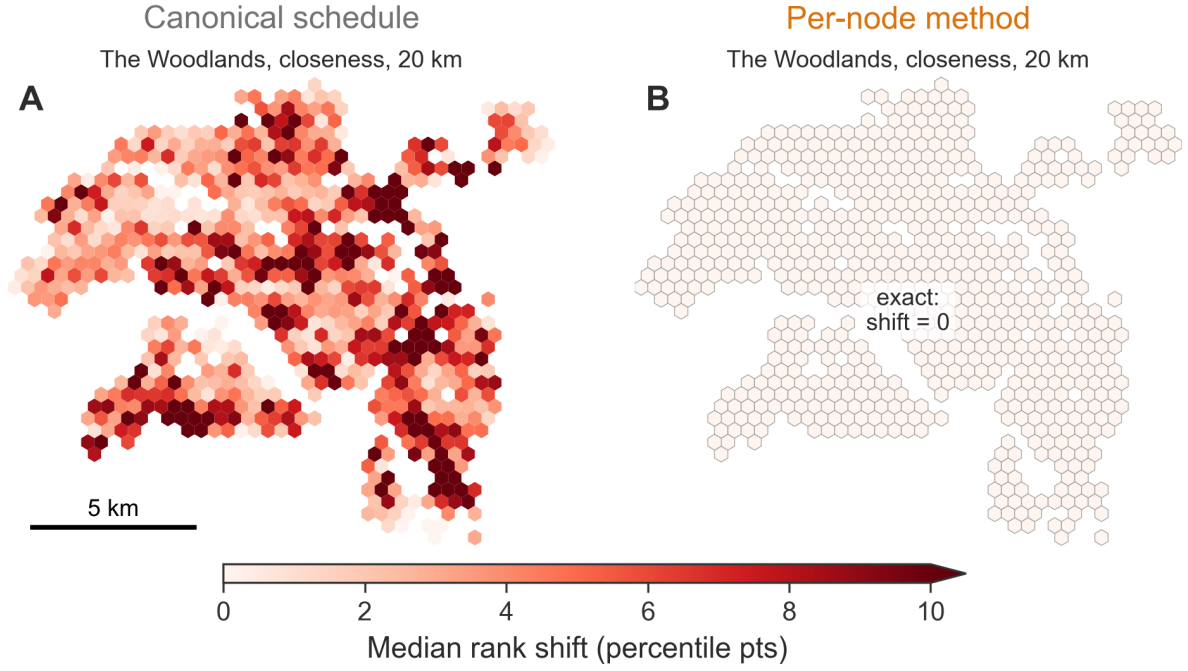


Figure 10: Closeness rank shift on the held-out network (The Woodlands) at 20 km under the canonical schedule (A) and the per-node method (B), drawn as hexbin medians on the fixed 0–10 percentile-point scale of Figure 7; cell medians above 10 saturate at the arrowed cap (the largest in panel A is 35 points). The comparison is between designs, not between two sampling runs: the schedule’s fixed rate samples this cell and its rank shifts cluster in space, whereas the per-node method’s work test recognises that sampling cannot pay here and routes the cell to exact computation, so panel B shows zero shift. That refusal is the method’s answer for this cell; the separate forced-sampling check (Section 6.3) confirms the sampled path would also meet the target. Betweenness is compared in Section 5.3: its improvement is reduced spatial clustering at similar magnitude, which a median map cannot show.